🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.AT.8

Equations and Inequalities Problem Solving

Lesson 7-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can model and solve real-world problems using equations and inequalities.
Language Objective: I can explain my reasoning using the words model, equation, inequality, and reasonableness.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Model | Equation | Inequality | Reasonableness | Variable | Constraint

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A drawing, equation, or diagram that represents a situation is a ______________.

2. A math sentence with an equal sign is an ______________.

3. A math sentence that compares values with <, >, ≤, or ≥ is an ______________.

4. Checking whether an answer makes sense for the problem is checking its ______________.

5. A letter that stands for the unknown amount in a problem is a ______________.

6 A limit that tells which values are allowed in a problem is a

________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What was your plan to solve this problem, and how did you check that your answer was reasonable?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Model** — To show a real-life situation with an equation or inequality.
  Modelar — Mostrar una situación real con una ecuación o desigualdad.
- [ ] **Equation** — A math sentence with an equal sign showing both sides are the same.
  Ecuación — Una oración matemática con un signo igual que muestra que ambos lados son iguales.
- [ ] **Inequality** — A math sentence that compares two sides with <, >, ≤, or ≥.
  Desigualdad — Una oración matemática que compara dos lados con <, >, ≤ o ≥.
- [ ] **Reasonableness** — Checking if your answer makes sense.
  Razonabilidad — Revisar si tu respuesta tiene sentido.
- [ ] **Variable** — A letter that stands for the unknown amount in a word problem.
  Variable — Una letra que representa la cantidad desconocida en un problema verbal.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

My plan was to ____, and I knew to ____ because ____.
Mi plan fue ____, y supe ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: An exact total points to an equation; a limit or range points to an inequality.

Evidence: Students use signal words ('equals/total' for an equation; 'at least/at most/no more than' for an inequality) to choose the correct model.

Reasoning: This evidence connects the definition of equation to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **My plan was to ____, and I knew to ____ because ____.**
  Mi plan fue ____, y supe ____ porque ____.
- **My answer is reasonable because ____.**
  Mi respuesta es razonable porque ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

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4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Model
2. **Notes 2:** Equation
3. **Notes 3:** Inequality
4. **Notes 4:** Reasonableness
5. **Notes 5:** Variable
6. **Notes 6:** Constraint
7. **Watch for this mistake:** *A common mistake in Equations and Inequalities Problem Solving is matching the wrong model to a key phrase. For 'the getaway car held no more than 6 bags,' students often write the equation $b = 6$ instead of the inequality $b \le 6$, because they see one number and assume it must be exact. The same mistake happens in reverse: for 'the reward split equally among 5 people gave each $40,' students sometimes write the inequality $5p \ge 40$ instead of the equation $5p = 40$, because 'split equally' sounds like it could mean a range. Before you submit, ask: does this phrase give one exact value (equation, =) or a limit/range (inequality, <, >, ≤, ≥)?*
8. **Try It 1:** A. $x \le 20$ — *'At most 20' means 20 or fewer, so $x \le 20$.*
9. **Try It 2:** A. $w = 12$ — *Divide both sides by 5: $w = 60 \div 5 = 12$ pounds. Check: $5 \times 12 = 60$ ✓*